Communicating with Data – Data Feminism

Dr. Ab Mosca (they/them)

Plan for today

- Power structures in data science
- Importance of context & documentation

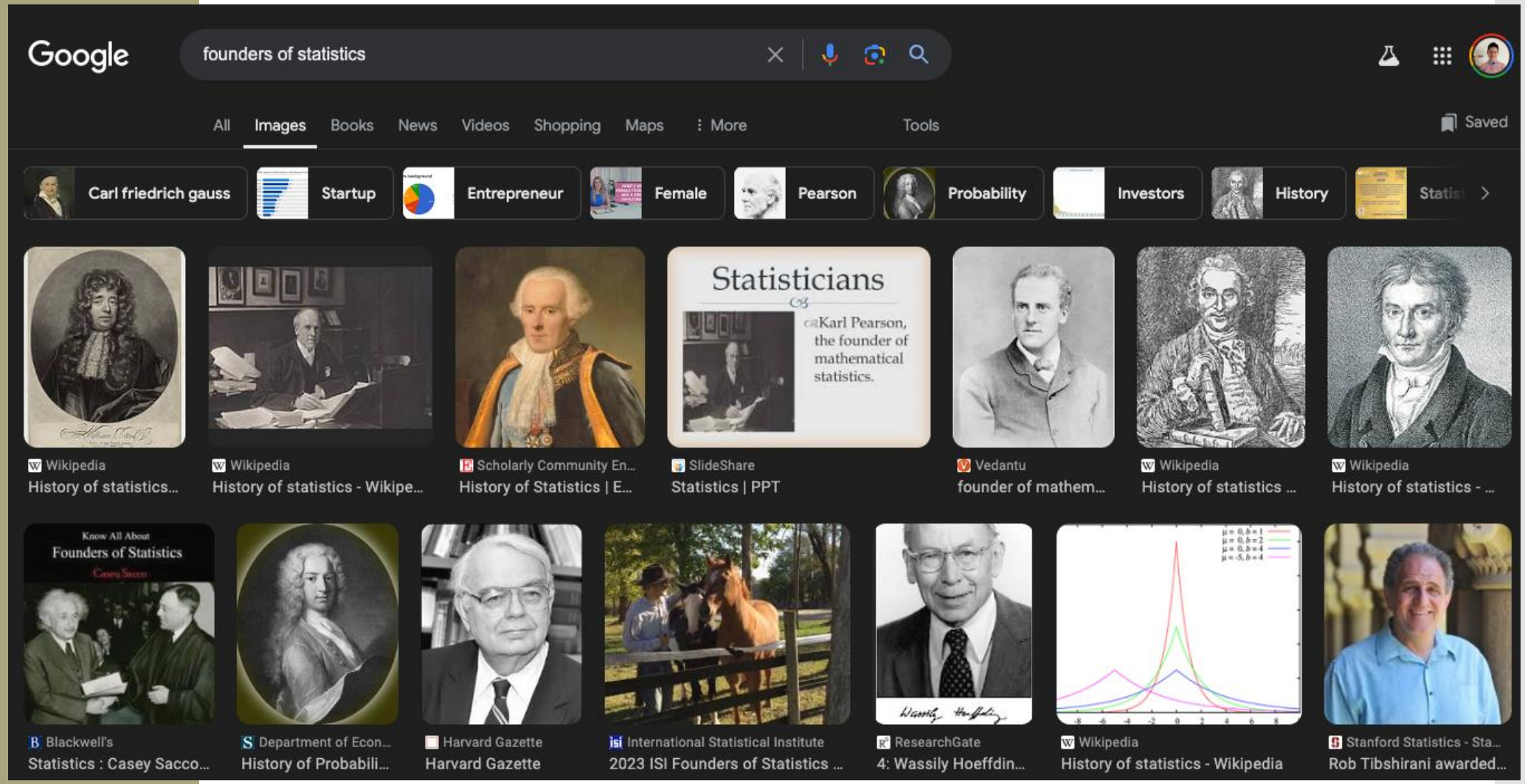
Discussion

What is the origin of data science?

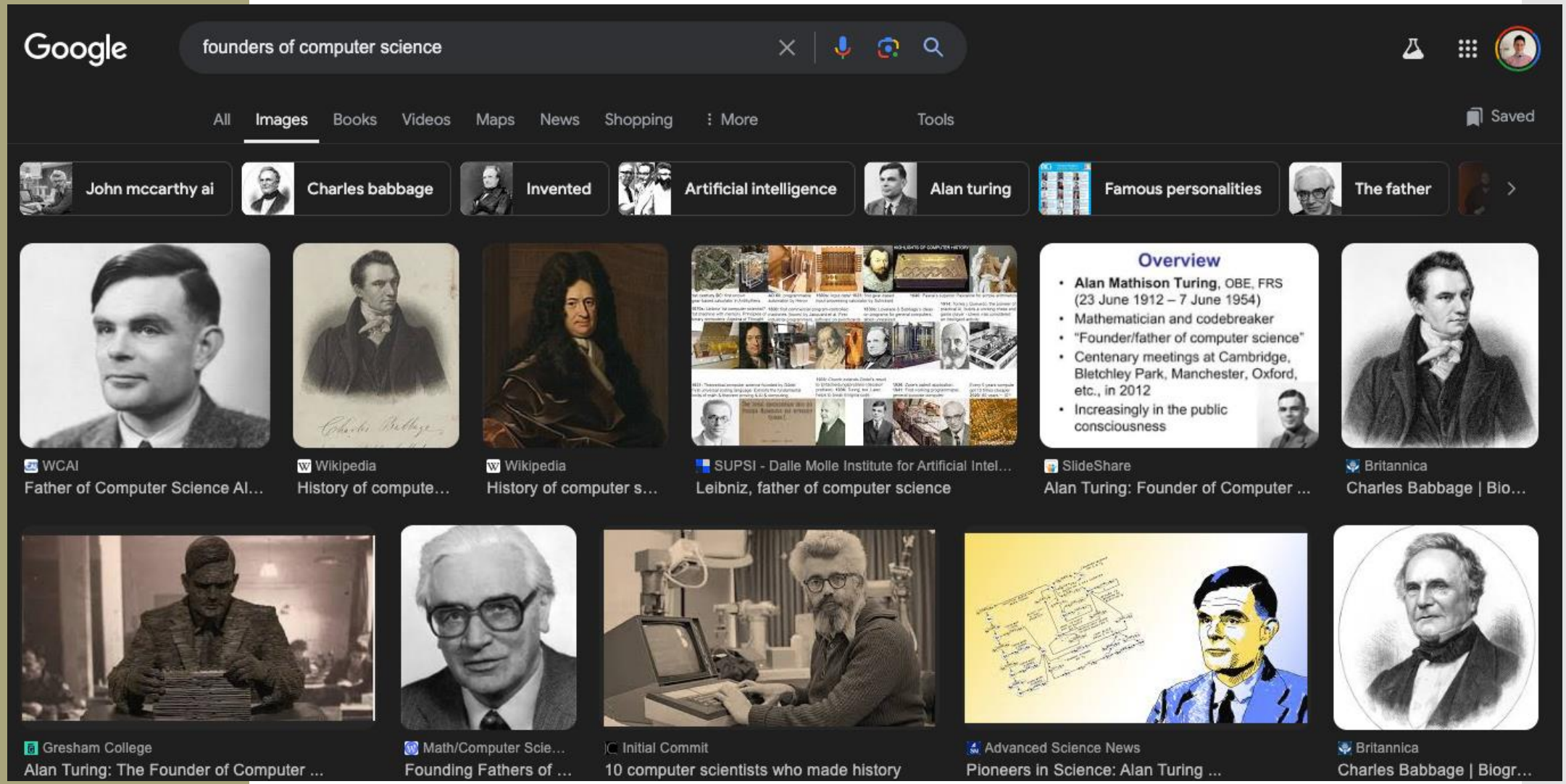
Where does it come from?

When did it start?

Who started it?



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Data Science & Power

- Hmmm….. Seems like data science is born from white cis males
  - Fun fact: this is not actually true if you dig deeper! But that's for another class

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Data Science & Power

Data Capitalism (Meyers West 2019)

- History is full of examples of data being used to control
- West argues, data as a commodity "enables an asymmetric redistribution of power that is weighted toward the actors who have access and the capability to make sense of information."

Data Science & Power

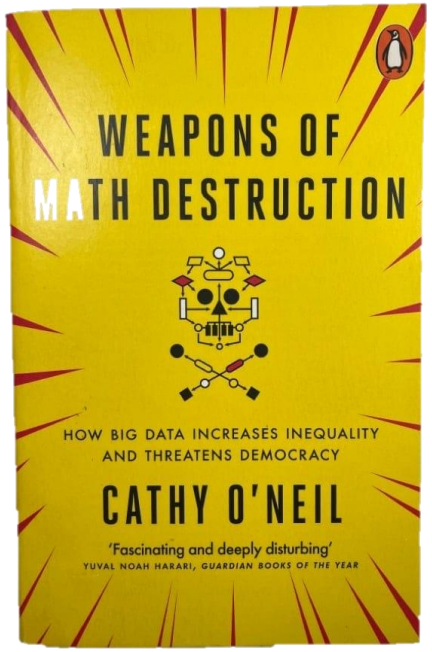
Data as Power

- In South American Andean cultures, Khipus are elaborate assemblages of knotted string used for millennia to record extracted numerical data such as tax records and military obligations of the populace (Medrano & Urton, 2018).
- From 2500BC the ancient Egyptian cultures were creating census datasets in order to determine how much labor force could be conscripted into the construction of pyramids for their pharaohs (Census-Taking in the Ancient World, 2016)

Data Science & Power

Algorithms as Power





Cathy O'Neil

https://www.youtube.com/watch?v=heQzqX35c9A

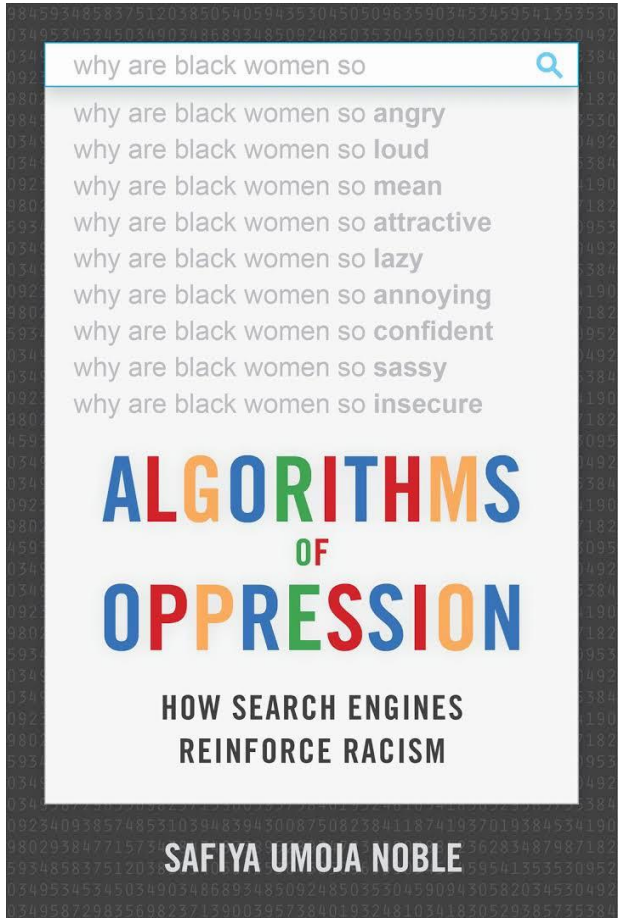
Data Science & Power

- The "Big Data" revolution argues that with enough data we can make unbiased decisions
- However, data science:
  - Lacks transparency
  - Employs extractive collection
  - Leverages technological complexity
  - Controls impact

Bhargava, R. (2023). Teaching Data That Matters: History and Practice

Example: Search Engines





Dr. Safiya Noble

https://youtu.be/iRVZozEEWlE?si=qzRtPmQzxl9KDxR2

Example: Search Engines

- Search engine algorithms are largely based on:
  - Profit
  - Historical data
  - Predictive analytics

What are downstream real-life impacts of this search engine bias?

Example: Facial Recognition

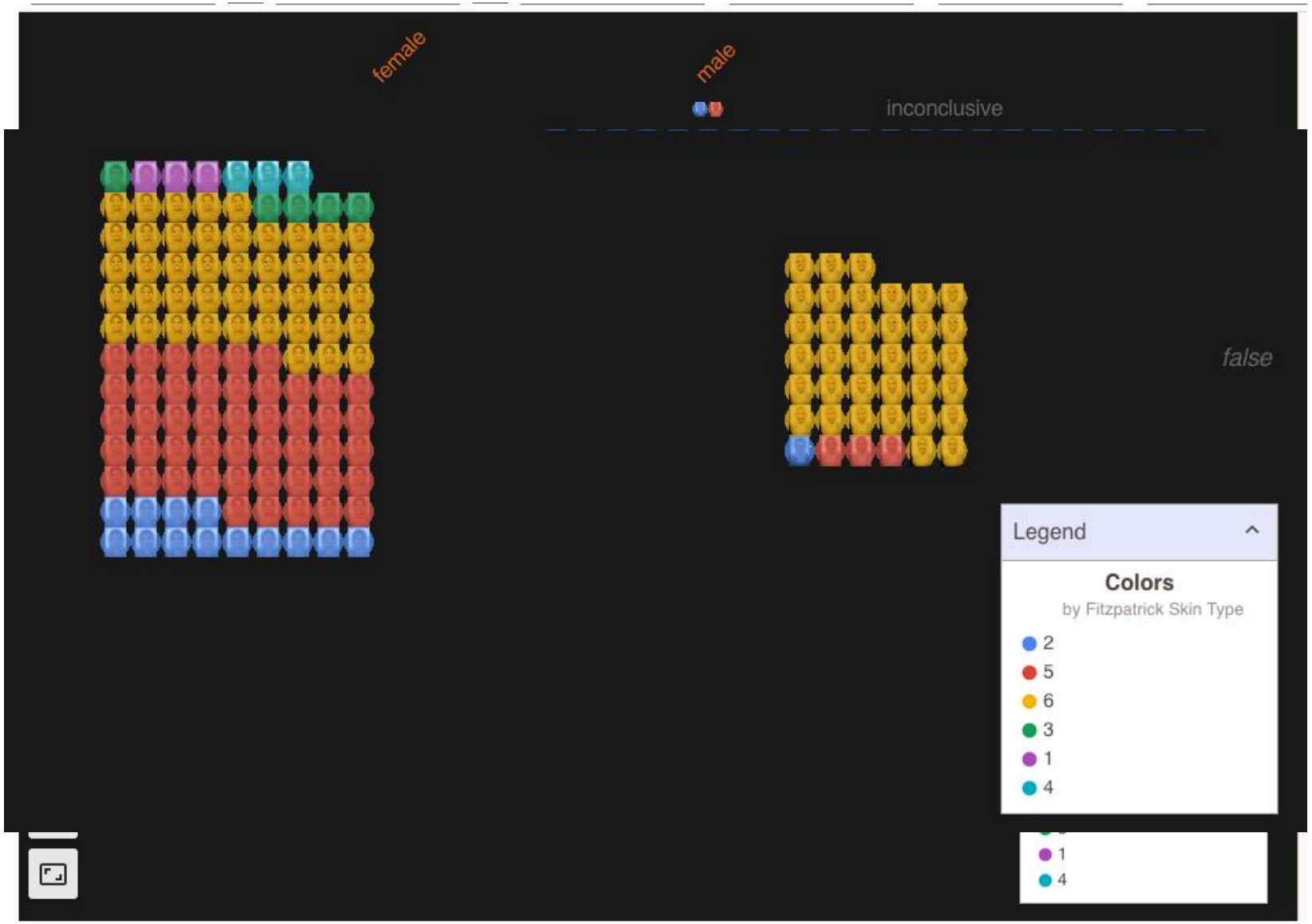


Dr. Joy Buolamwini

https://youtu.be/UG_X_7g63rY?si=qDMmUX5VjpaJYURe

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Example: Facial Recognition



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Example: Facial Recognition

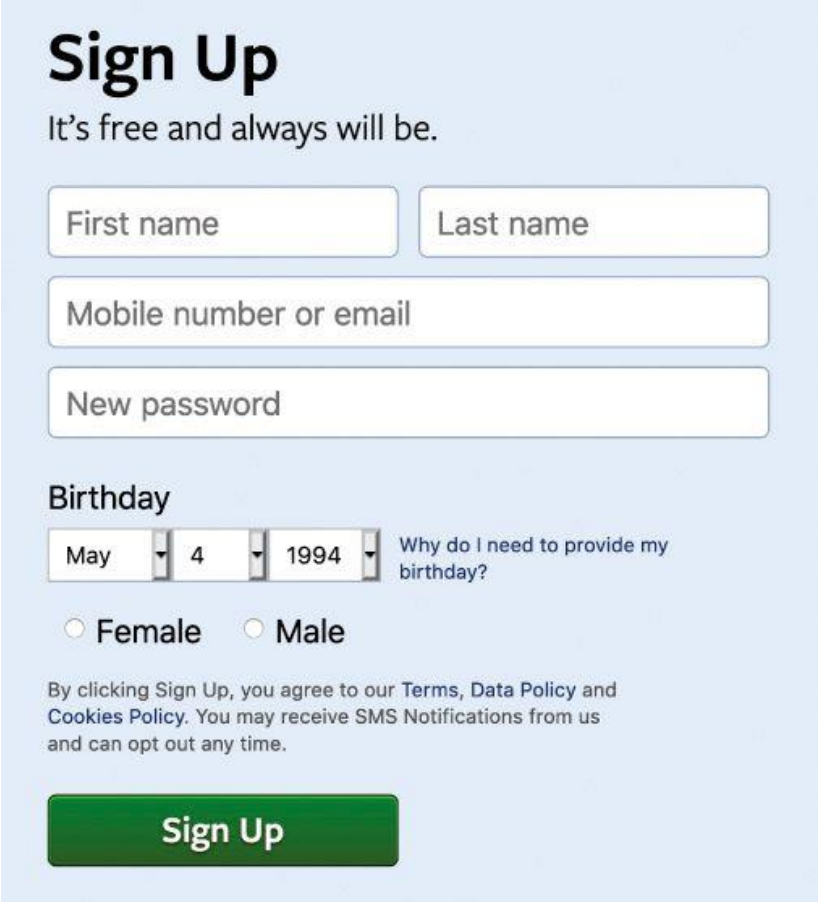
- Training dataset used for most facial recognition systems contains
  - 78% male faces
  - 84% white faces
  - Only 4% were women and dark-skinned

What are downstream real-life impacts of this algorithmic bias?

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"What gets counted counts"

- Data is often used to inform policy and allocate resources
- What is not counted in that data collection can become invisible
  - Ex. Expansive gender



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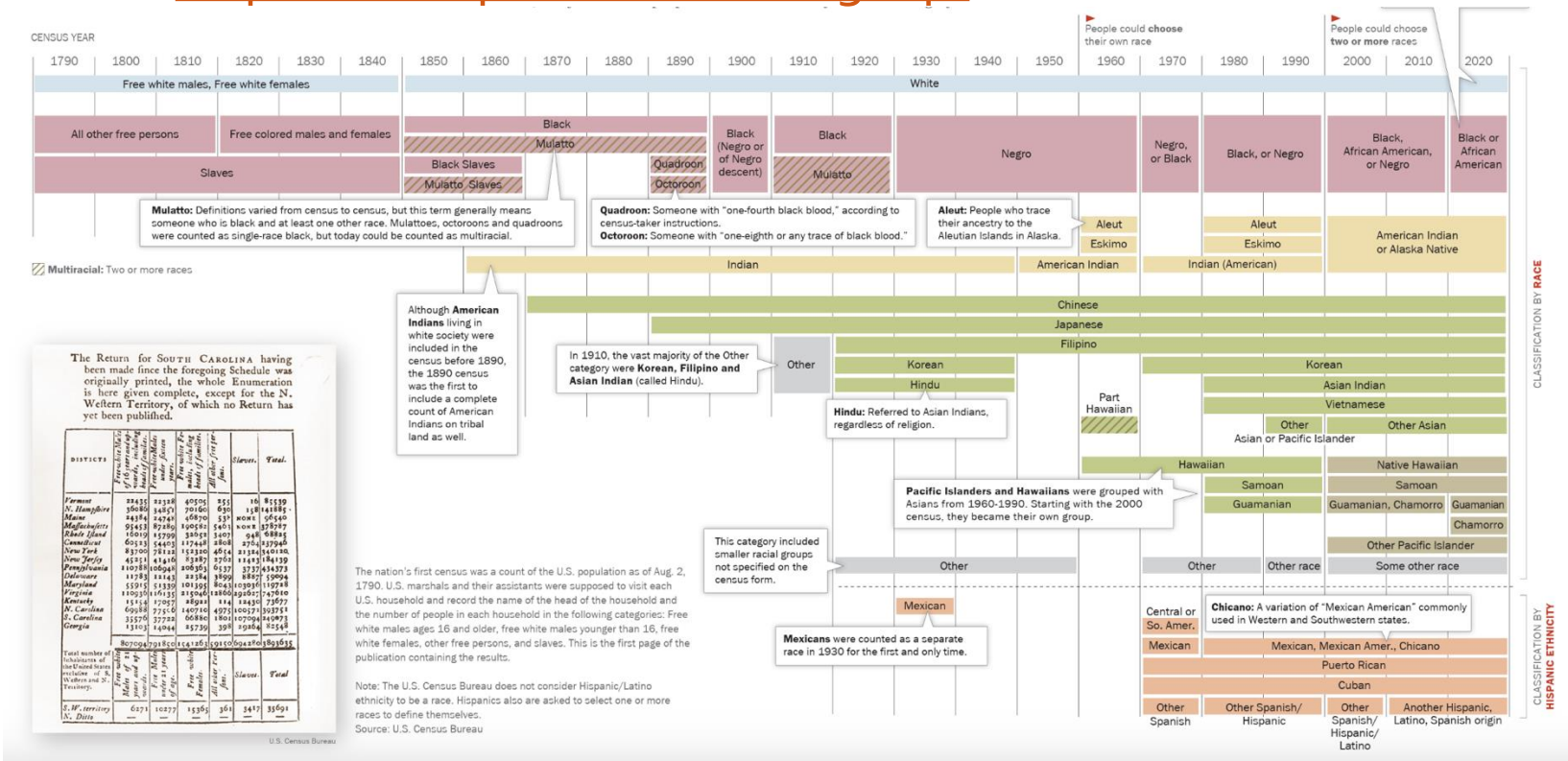
"What gets counted counts"

- Data is often used to inform policy and allocate resources
- What is not counted in that data collection can become invisible
  - Ex. US Census



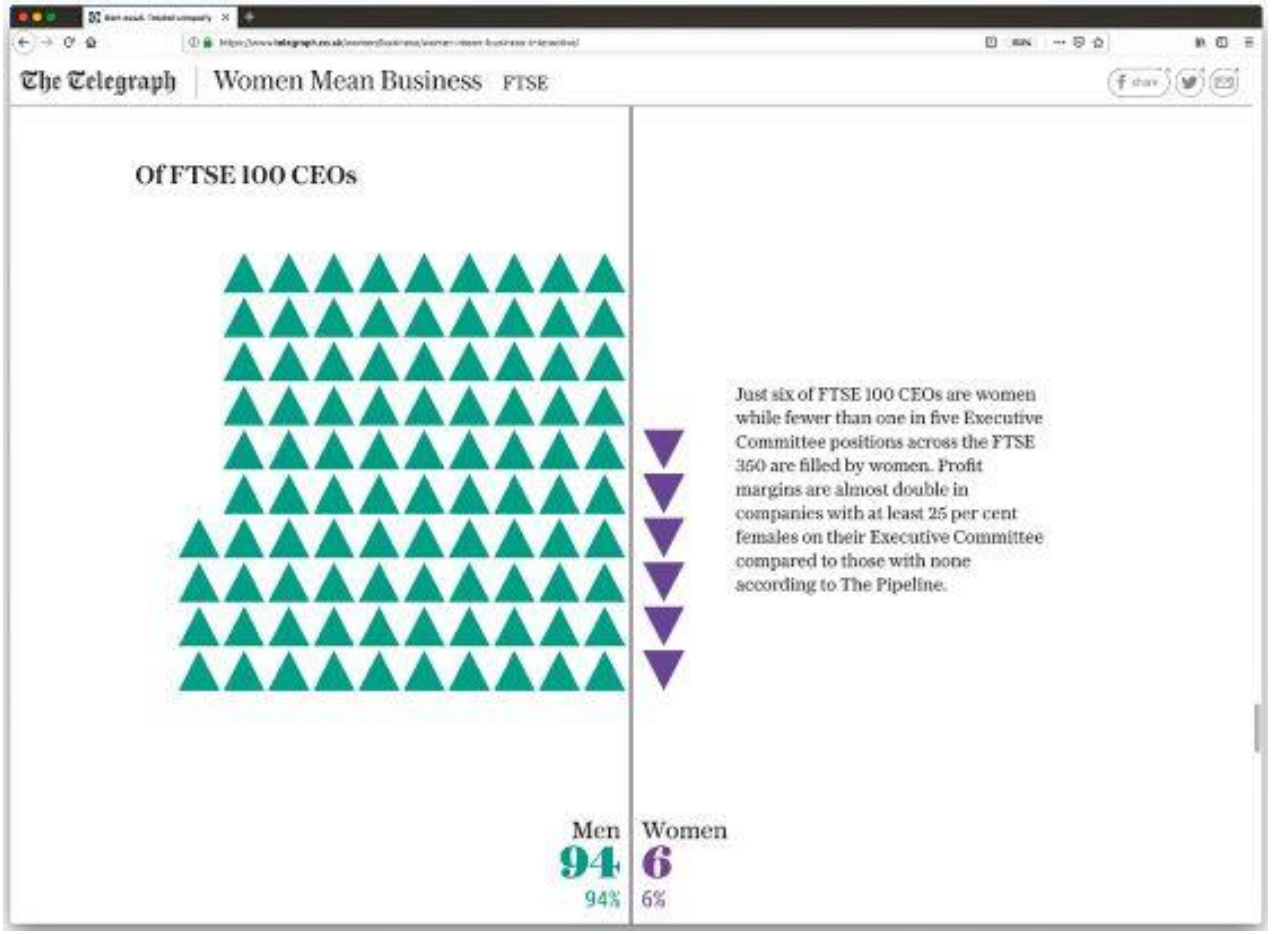
"What gets counted counts"

- Data is often used to inform policy and allocate resources
- What ***is*** counted is considered important
  - Ex. US Census & Race
  - https://www.pewresearch.org/social-trends/feature/what-census-calls-us/
  - https://www.pewresearch.org/wp-



Subverting Power

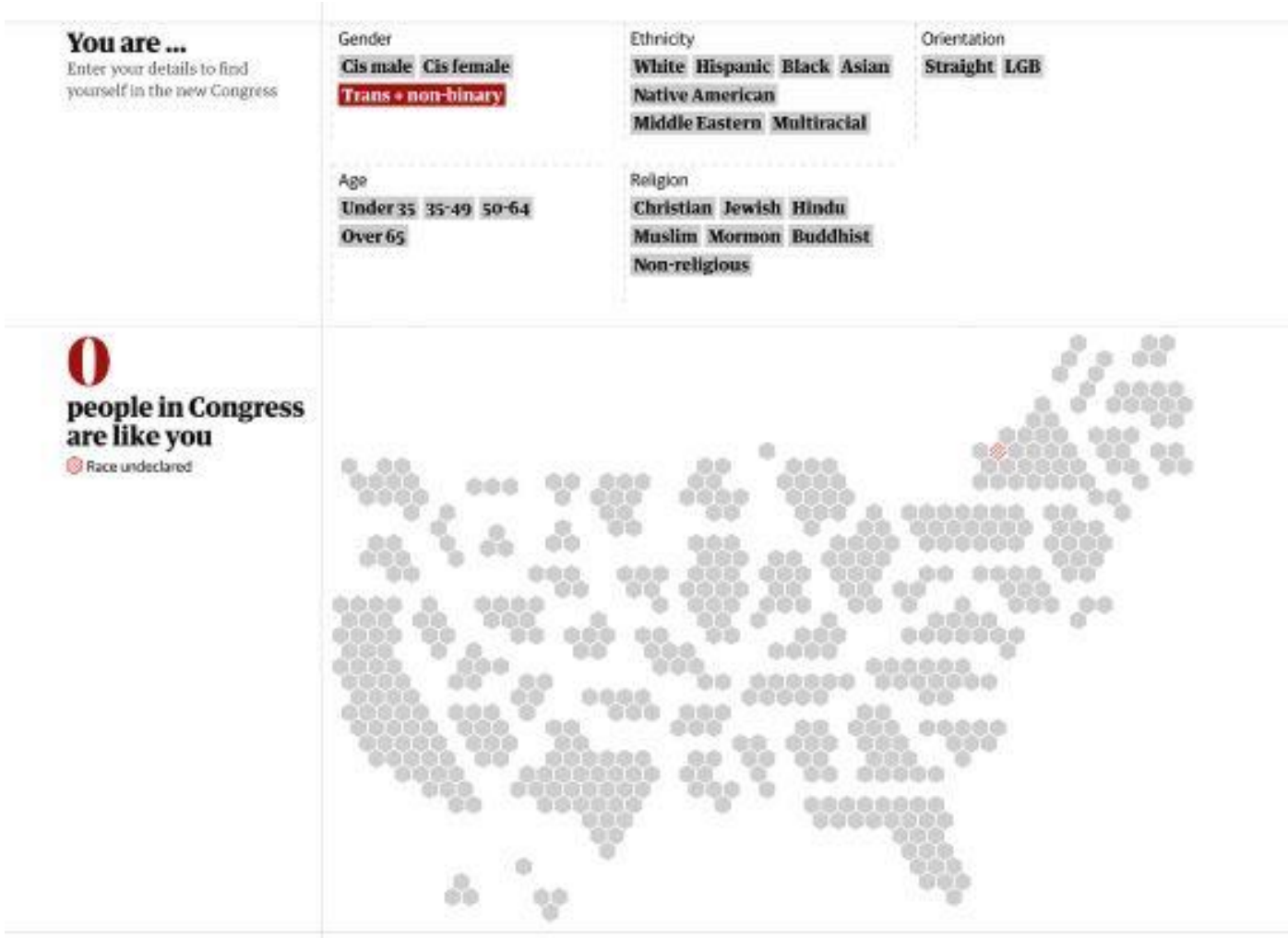
- Subvert norms



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Subverting Power

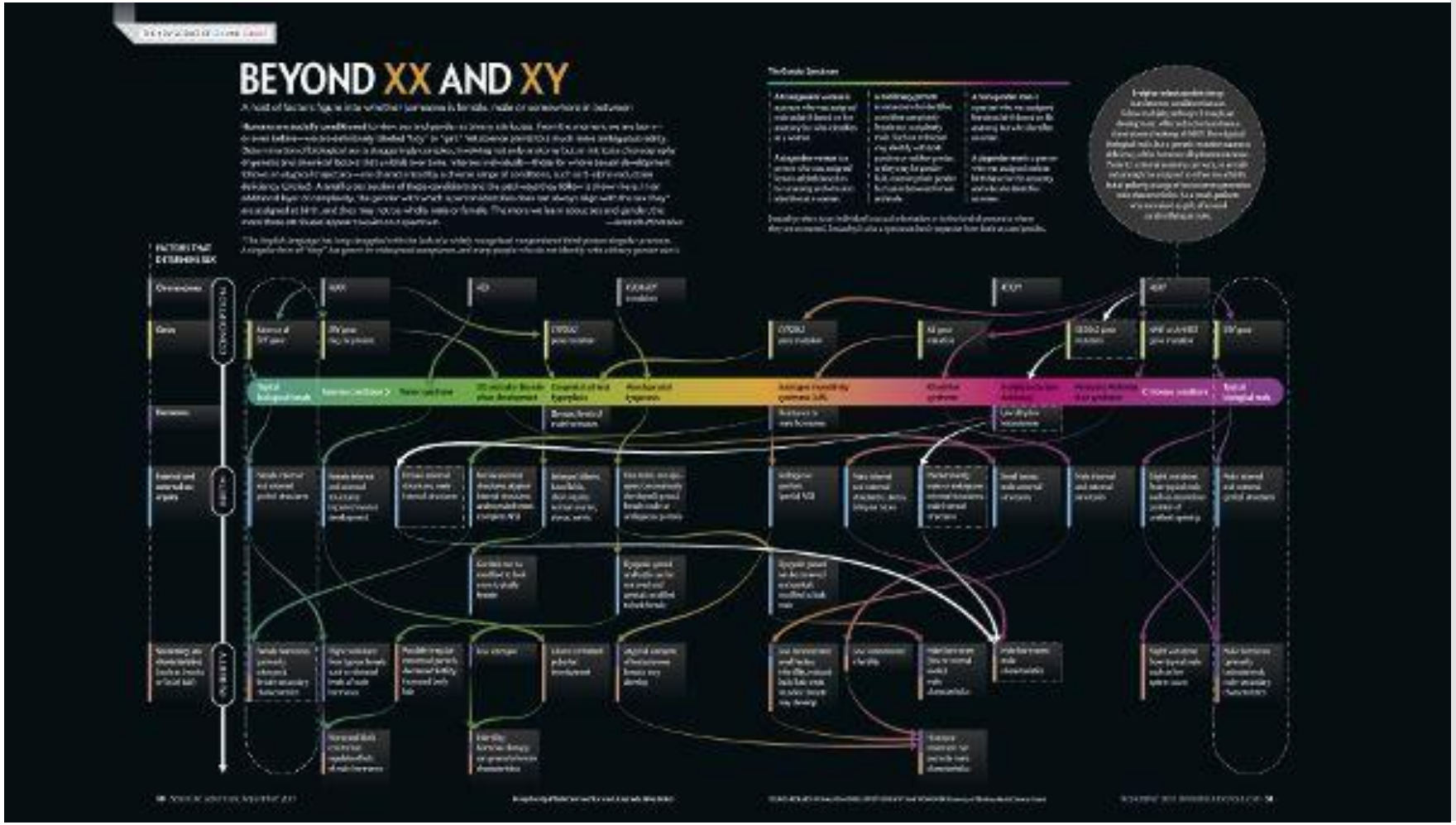
- Highlight missing categories



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Subverting Power

- Educate



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Subverting Power

- Rethink data collection



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Subverting Power

Table 5.1: Features of "data for good" versus data for co-liberation

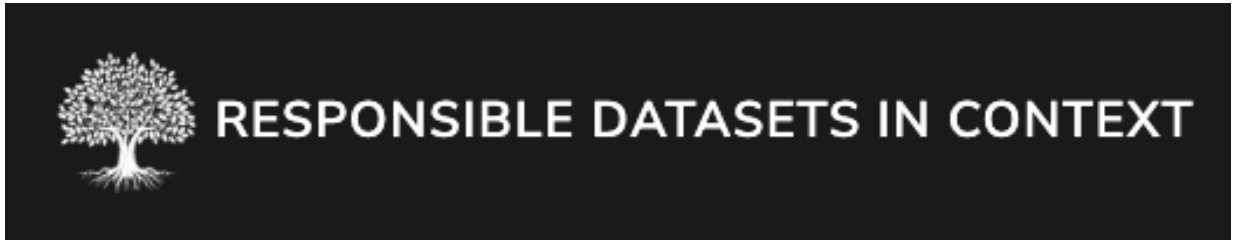
	"Data for good"	Data for co-liberation
Leadership by members of minoritized groups working in community		√
Money and resources managed by members of minoritized groups		√
Data owned and governed by the community		√
Quantitative data analysis "ground truthed" through a participatory, community-centered data analysis process		√
Data scientists are not rock stars and wizards, but rather facilitators and guides		√
Data education and knowledge transfer are part of the project design		√
Building social infrastructure—community solidarity and shared understanding—is part of the project design		√

- Add transparency
- Avoid extractive approaches
- Follow the lead of the community

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Acknowledge context

Subverting Power



https://www.responsible-datasets-in-context.com/datasets.html

- What is the historical context of the data?
- Where did the data come from? Who collected it?
- Why was the data collected?
- How was the data collected?
- How is the data used?
- What's in the data?
- What "counts" as a data point?
- What data is missing?
- How is uncertainty handled?

What biases or ethical issues could the answers to these questions reveal that would otherwise be hidden?

Your Turn!

- Find 1-2 people to work with
- Below a report is linked. Your job is to investigate the data behind the report. To do this identify:
  - The historical context of the data
  - Where the data came from / Who collected the data
  - Why the data was collected
  - How the data was collected
  - What is in the data
  - What "counts" as a data point
  - What data is missing
  - How uncertainty was handled in data collection

https://www.mass.edu/datacenter/PMRS/holyoke.asp